

NICKOLAS BOYLAN

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EDUCATION

Tufts University, Medford, MA

School of Engineering

Bachelor of Science in Computer Science (BSCS)

09/2021 - 05/2025

GPA: 4.0

Relevant Coursework: *Web Programming, Cybersecurity, Parallel Programming, and Software Engineering*

TECHNICAL SKILLS

- Fullstack Programming
- Machine Learning
- Reinforcement Learning
- Multithreaded Programming
- Cloud Web Hosting
- General Cybersecurity
- Network Security
- Testing and Debugging

Programming Languages:

C/C++ | C# | Java | Python | HTML | CSS |
Javascript | R | PHP | Scheme | Erlang |
Assembly

Frameworks/Tools:

MySQL | MSSQL | MongoDB | .NET | CUDA
| Git + GitHub | PyBullet | ROS | PyTorch |
Heroku | Node.js | SiteGround | Unix | React

GENERAL SKILLS

- Communication and Collaboration
- Leadership
- Time Management
- Multitasking
- Adaptability
- Attention to Detail
- Customer Service
- Microsoft Office Products

LANGUAGES

- English
- French
- Japanese

SUMMARY

Recent software engineering graduate with a diverse array of technical skills including a strong foundation in multithreaded programming. A quick learner and eager to contribute in a collaborative environment.

EXPERIENCE

Multimodal Learning, Interaction, and Perception (MULIP) Lab – Tufts University **RL Trained Soccer Bot Project** (09/2024 – 05/2025)

- Designed and implemented an architecture to train and test a reinforcement learning agent in a PyBullet based simulation to play soccer against an opponent
- Final model incorporated a hierarchical policy that swapped between separately trained offense and defense policies
- Optimized training algorithm to train 10 million timesteps 60% faster
- Experimented training agents on various reward functions, both dense and sparse, to optimize the agent's performance
- Added support for recording gameplay trajectories and pretraining the policy/model on those trajectories

UNIVERSITY PROJECTS

Live Concurrent Web Scraper (01/2025 – 04/2025)

- Created a fullstack web application to automate the scraping of drug development pipeline data from 10+ major pharmaceutical companies
- Implemented a custom backend capable of concurrently receiving and handling requests to scrape, ensuring high responsiveness even under heavy load
- Designed an intuitive frontend UI for filtering, sorting, and exporting the scraped data sent from the backend

Dynamic UI Color Palette Generator (01/2024 – 04/2024)

- Developed a web application to generate and preview a custom color palette for website designs in real time
- Designed a clean and easy to use frontend that dynamically interacts with the backend to provide personalized content
- Integrated an SQL database to store user generated palettes for future access
- Deployed the app on a cloud-hosted server to process requests and allow communication between the client and server

OTHER EXPERIENCE

[June 2020 - Present] **Boylan Properties Inc**, Remote - *Accounting Bookkeeper*

- Create and organize financial reports. Verify accuracy of financial records

[Sep 2022 - May 2025] **Tufts Academic Space Management**, Tufts Campus - *Receptionist*

- Served guests. Attended phone call/email requests. Monitored building condition

[Jan 2022 - May 2022] **Tufts Dining Services**, Carmichael Hall - *Dining Assistant*

- Ensured clean and organized facilities. Provided service to guests